

POCKETSMART AI

Your Smart Budget & Recommendation Assistant

TEAM DETAILS

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1. Abstract

PocketSmart AI is a smart budgeting and recommendation system designed to help users manage their personal expenses effectively. The system allows users to enter their income, expenses, and spending categories. Based on the entered information, the system analyzes spending patterns and provides simple recommendations for better money management.

The main purpose of PocketSmart AI is to make budgeting easy for students and other users. It can help users understand where their money is being spent, set spending limits, and receive suggestions for saving money.

2. Introduction

Managing money can be difficult when people do not keep track of their daily expenses. Small expenses can add up and make it difficult to understand where the money is going.

PocketSmart AI solves this problem by providing a simple digital assistant for budgeting. Users can enter their income and expenses, and the system organizes the information into different categories such as food, travel, shopping, education, and entertainment.

Using basic AI-based analysis, the system can identify spending patterns and provide personalized recommendations.

3. Problem Statement

Many students and individuals spend money without properly tracking their expenses. Existing budgeting methods may be difficult to maintain manually.

The problems include:

- Difficulty in tracking daily expenses.
- Lack of proper monthly budgeting.
- Unnecessary spending.
- Difficulty identifying high-spending categories.
- Lack of personalized saving suggestions.

Therefore, a simple and intelligent budgeting assistant is needed.

4. Objectives

The main objectives of PocketSmart AI are:

1. To record and categorize user expenses.
2. To calculate total income and expenses.
3. To track the user's spending pattern.
4. To help users create a budget.
5. To identify unnecessary or high spending.
6. To provide personalized recommendations.
7. To encourage better saving habits.
8. To present financial information in a simple and understandable way.

5. Proposed System

PocketSmart AI is proposed as a user-friendly budgeting application.

The user enters details such as monthly income, expense amount, expense category, date of expense, and budget limit.

The system processes this information and displays total income, total expenses, remaining balance, category-wise expenses, budget status, spending analysis, and saving recommendations.

6. Key Features

6.1 Expense Tracking: Users can enter their daily expenses and select the appropriate category.

6.2 Budget Management: Users can set a monthly budget and monitor their spending.

6.3 Expense Categorization: Expenses can be categorized into Food, Travel, Shopping, Education, Entertainment, Bills, and Other.

6.4 Spending Analysis: The system analyzes the user's expenses and identifies categories where the user spends more money.

6.5 Smart Recommendations: PocketSmart AI can provide suggestions such as reducing unnecessary spending and setting limits for high-expense categories.

6.6 Dashboard: A simple dashboard can display income, expenses, savings, and budget status.

7. System Workflow

User Registration/Login → Enter Income → Add Expenses → Categorize Expenses → Analyze Spending → Compare With Budget → Generate Recommendations → Display Results

8. Technologies Used

Frontend: HTML, CSS, JavaScript

Backend: Python

Database: SQLite / MySQL

AI/Data Analysis: Python, Pandas, NumPy, and basic machine-learning or rule-based recommendation techniques

Development Environment: Visual Studio Code, Python

9. System Modules

Module 1: User Management — Manages basic user information and login details.

Module 2: Income Management — Users can enter and update their income information.

Module 3: Expense Management — Users can add, edit, and delete expenses.

Module 4: Budget Management — Users can create spending limits for different categories.

Module 5: Expense Analysis — Calculates total expenses and analyzes category-wise spending.

Module 6: Recommendation System — Generates simple recommendations based on spending patterns.

Module 7: Dashboard — Displays important information in an easy-to-understand format.

10. AI Recommendation Logic

The recommendation system can use the user's spending information to generate suggestions.

Example:

Monthly income: ■20,000

Food expenses: ■6,000

Travel expenses: ■2,000

Shopping expenses: ■5,000

The system identifies food and shopping as major expense categories and may provide a suggestion such as: “Your spending on food is relatively high. Consider setting a monthly food budget.”

The recommendation can be improved later using machine-learning techniques based on previous spending history.

11. Advantages

- Easy to use.
- Helps track daily expenses.
- Provides better understanding of spending habits.
- Helps users maintain a budget.
- Provides personalized suggestions.
- Reduces the need for manual calculations.
- Useful for students and individuals.

12. Limitations

- Recommendations depend on the information entered by the user.
- It does not replace professional financial advice.
- Incorrect expense data may produce inaccurate recommendations.
- Advanced AI features require sufficient historical data.

13. Future Enhancements

In the future, PocketSmart AI can be improved by adding:

- AI chatbot for financial queries.
- Voice-based expense entry.
- Automatic expense categorization.
- Spending prediction.
- Monthly financial reports.
- Graphs and charts.
- Bill and reminder notifications.
- Personalized saving goals.
- Mobile application.

- More advanced machine-learning recommendations.

14. Expected Result

The expected result of PocketSmart AI is a simple and intelligent budgeting assistant that helps users understand their financial habits.

The system should allow users to see their income, expenses, remaining balance, category-wise spending, and personalized recommendations in one place.

15. Conclusion

PocketSmart AI: Your Smart Budget & Recommendation Assistant is a useful project that combines budgeting, data analysis, and artificial intelligence.

The project helps users track their expenses, manage their budgets, identify spending patterns, and receive useful recommendations. It is especially useful for students who want to develop better money-management habits.

With future improvements such as machine learning, voice input, automatic categorization, and a mobile application, PocketSmart AI can become a more advanced personal budgeting assistant.

16. References

1. Python Documentation
2. Pandas Documentation
3. NumPy Documentation
4. SQLite Documentation
5. Basic concepts of Artificial Intelligence and Machine Learning